Employee Database Management System using MySQL

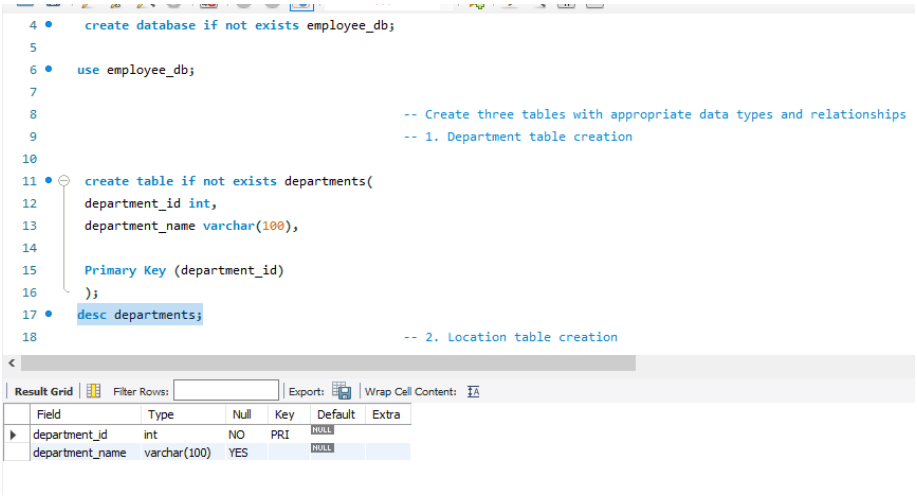
Data Pre-Processing (MySQL)

Tasks Performed:

DDL Commands:

1. Database and Table Creation (CREATE):

Department table creation



```
create database if not exists employee_db;

use employee_db;

                                        -- Create three tables with appropriate data types and relationships
                                        -- 1. Department table creation

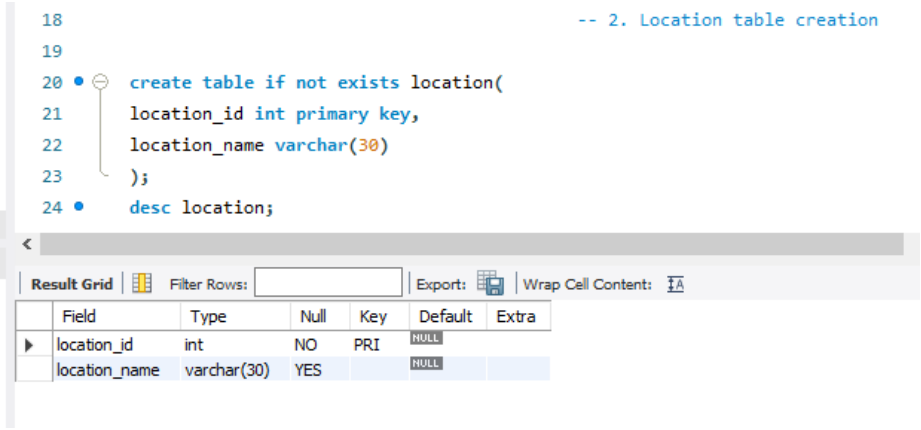
create table if not exists departments(
department_id int,
department_name varchar(100),

Primary Key (department_id)
);
desc departments;
                                        -- 2. Location table creation
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI	NULL	
department_name	varchar(100)	YES		NULL	

Location table creation



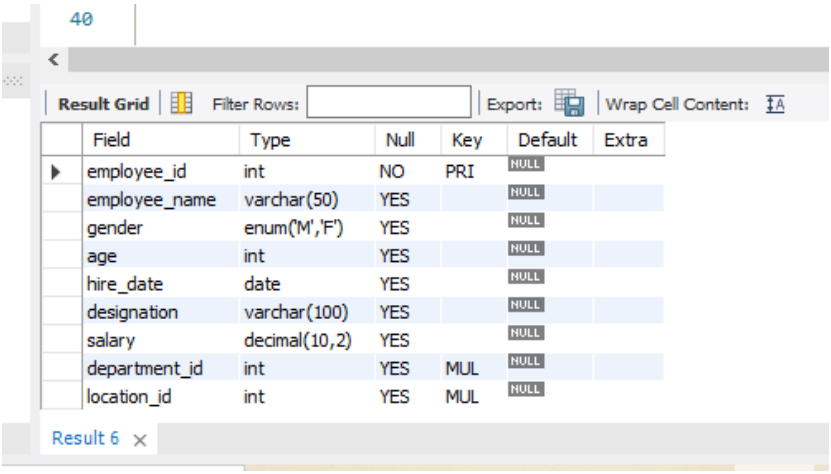
```
                                        -- 2. Location table creation

create table if not exists location(
location_id int primary key,
location_name varchar(30)
);
desc location;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
location_id	int	NO	PRI	NULL	
location_name	varchar(30)	YES		NULL	

Employees table creation



40

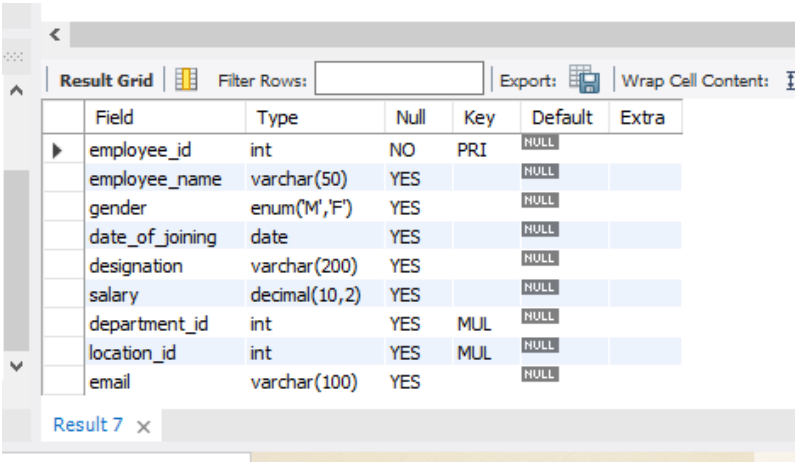
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	YES		NULL	
gender	enum('M','F')	YES		NULL	
age	int	YES		NULL	
hire_date	date	YES		NULL	
designation	varchar(100)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	

Result 6

2. Table Alteration (ALTER)

- Add a new column named "email" to the Employees table to store employee email addresses.
- Modify the data type of the "designation" column in the Employees table to support a wider range of values.
- Drop the “age” column from the Employees table.
- Rename the “hire_date” column to “date_of_joining”



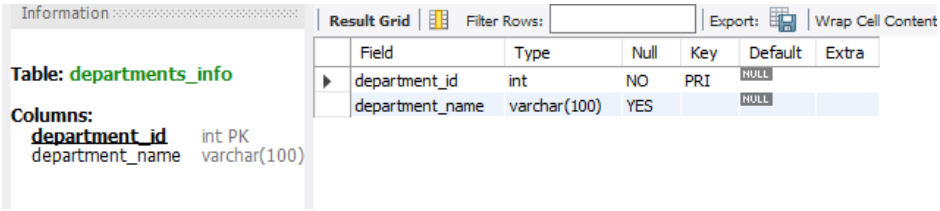
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

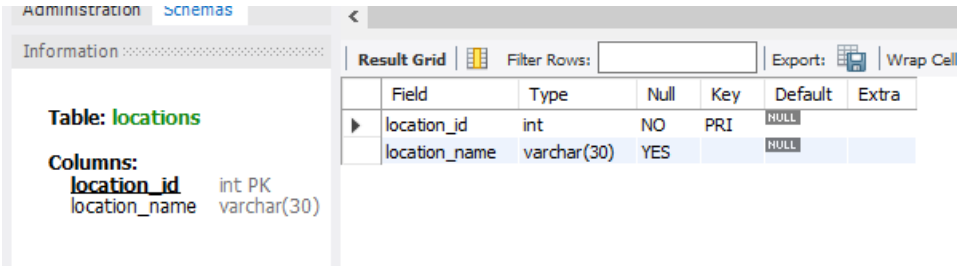
Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	YES		NULL	
gender	enum('M','F')	YES		NULL	
date_of_joining	date	YES		NULL	
designation	varchar(200)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	
email	varchar(100)	YES		NULL	

Result 7

3. Table Renaming (RENAME):

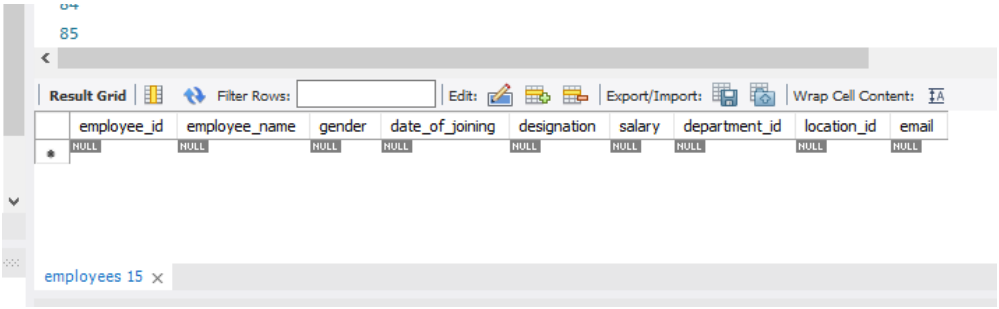
- Rename the "Departments" table to "Departments_Info"
- Rename the "Location" table to "Locations





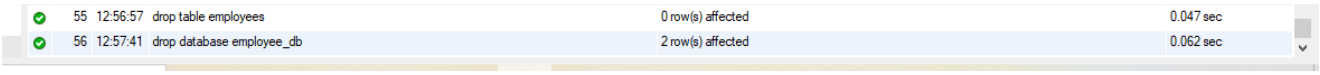
4. Table Truncation (TRUNCATE):

- Truncate the Employees table



5. Database & Table Dropping (DROP):

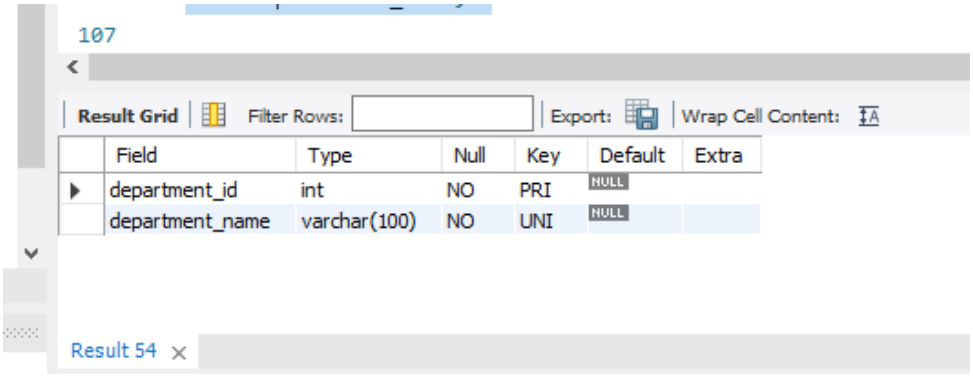
- Drop the Employees table and then the “employee” database



Constraints:

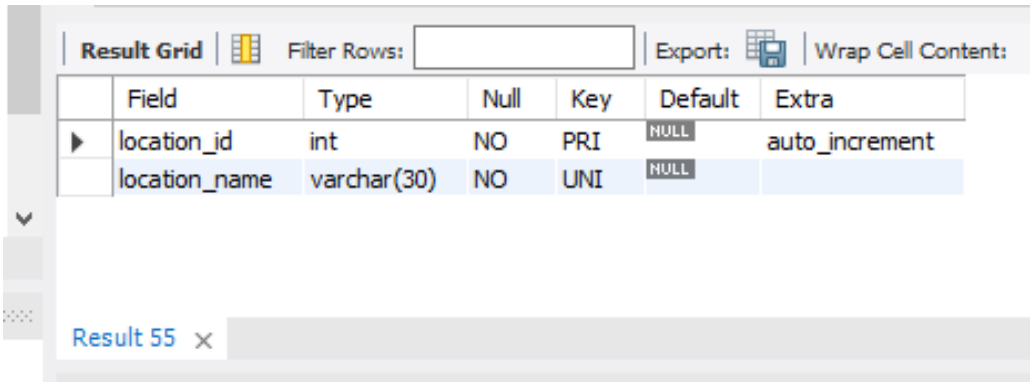
1.Departments Table:

- Ensure that the "department_id" uniquely identifies each department
- Set up constraints on the "department_name" to avoid duplicate and null entries



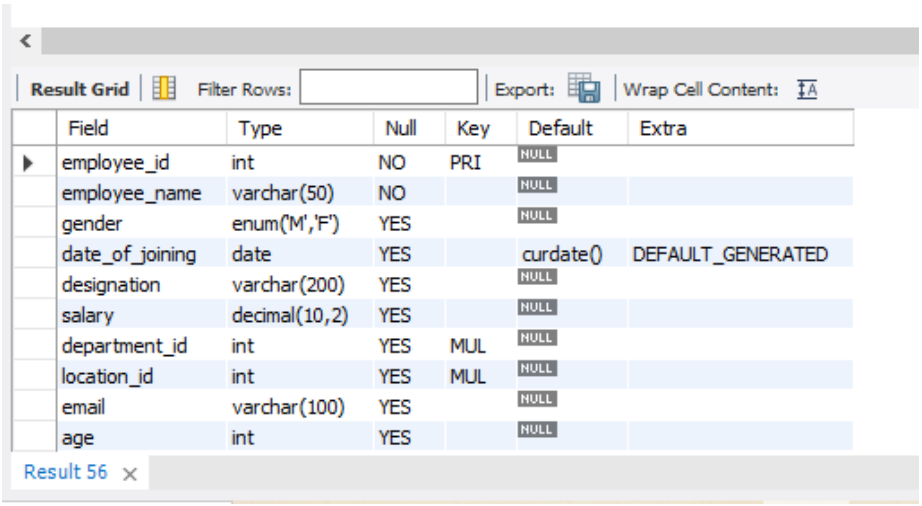
2.Locations Table:

- create incremented sequential for location_id
- Set up constraints on the "location_name" to avoid null and duplicate locations



3.Employees Table:

- Guarantee that each employee has a distinct identifier.
- Create a restriction to ensure that the employee's name is always provided.
- Limit the acceptable values for the "gender" field to only 'M' or 'F'.
- Enforce a condition to ensure that the employee's age is 18 or above.
- Automatically assign the current date to the "hire_date" field if not specified



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

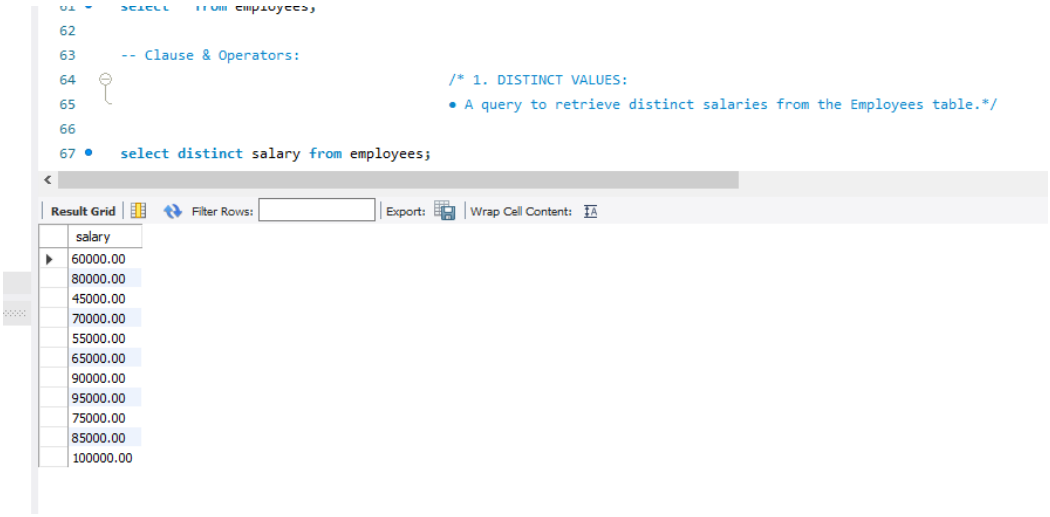
Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	NO		NULL	
gender	enum('M','F')	YES		NULL	
date_of_joining	date	YES		curdate()	DEFAULT_GENERATED
designation	varchar(200)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	
email	varchar(100)	YES		NULL	
age	int	YES		NULL	

Result 56

Clause & Operators:

1. DISTINCT VALUES:

● A query to retrieve distinct salaries from the Employees table



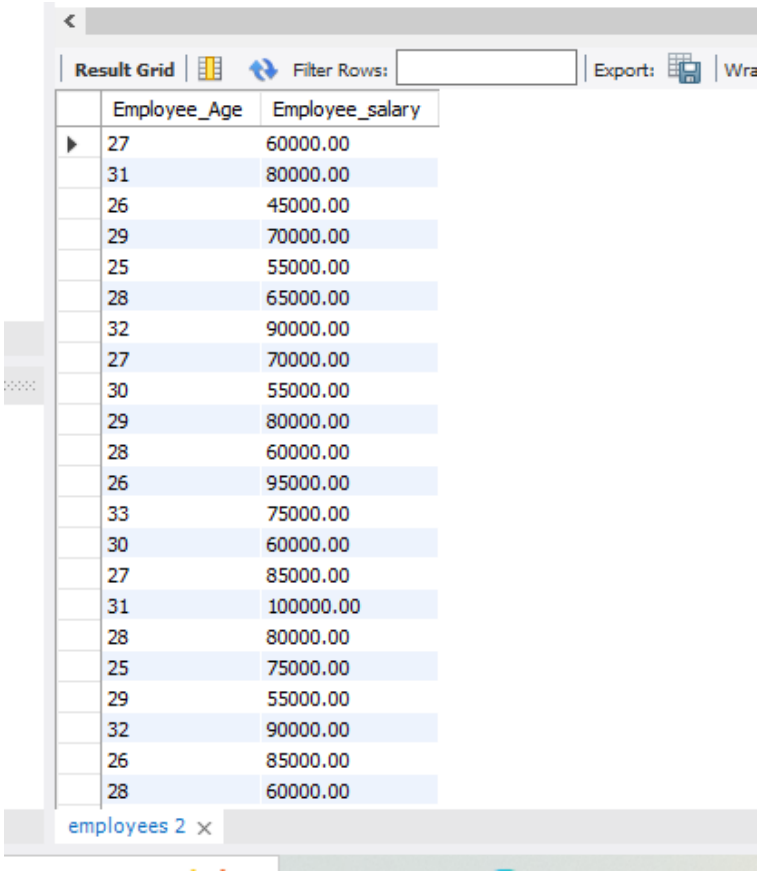
```
62
63      -- Clause & Operators:
64                                          /* 1. DISTINCT VALUES:
65                                          • A query to retrieve distinct salaries from the Employees table.*/
66
67 •    select distinct salary from employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

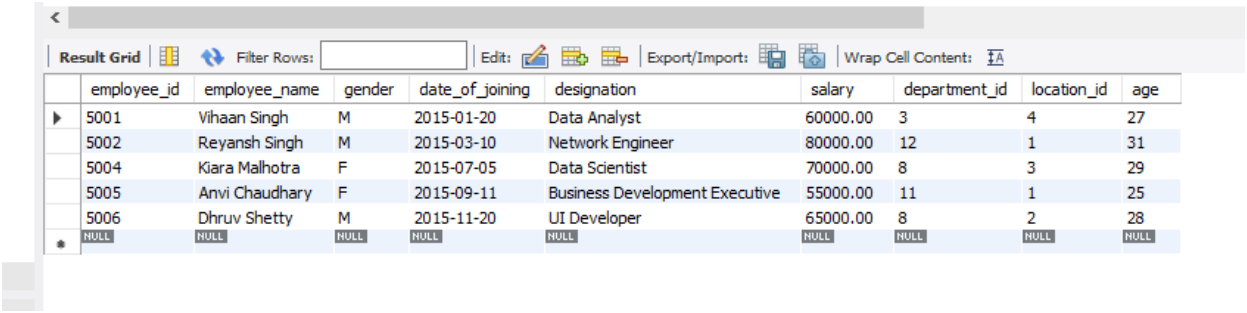
2. ALIAS (AS):

● Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively



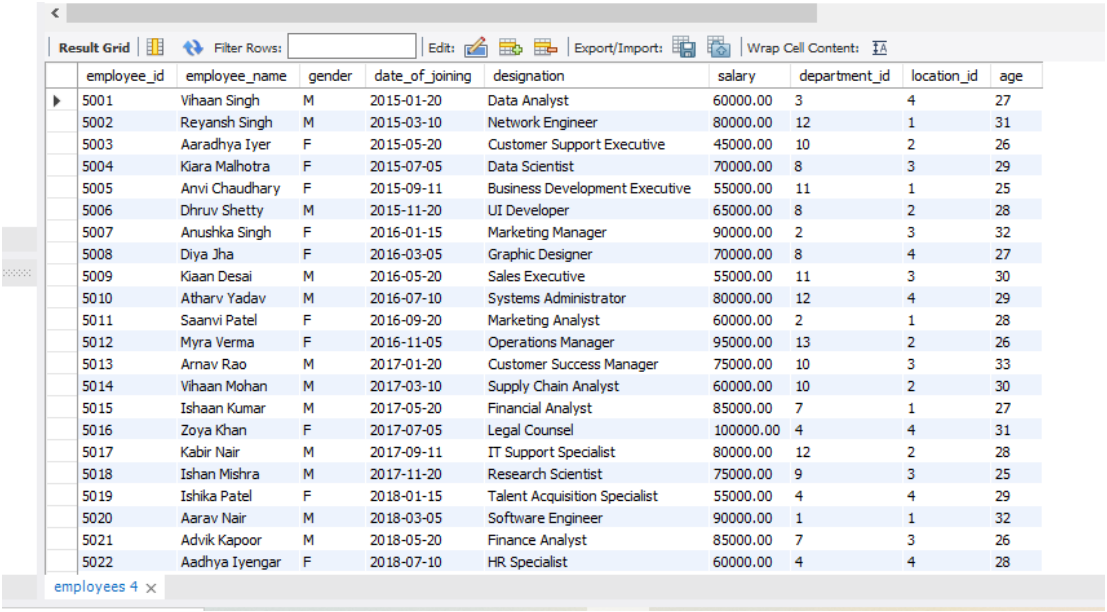
3. WHERE CLAUSE & OPERATORS:

● Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.



employee_id	employee_name	gender	date_of_joining	designation	salary	department_id	location_id	age
5001	Vihaan Singh	M	2015-01-20	Data Analyst	60000.00	3	4	27
5002	Reyansh Singh	M	2015-03-10	Network Engineer	80000.00	12	1	31
5004	Kiara Malhotra	F	2015-07-05	Data Scientist	70000.00	8	3	29
5005	Anvi Chaudhary	F	2015-09-11	Business Development Executive	55000.00	11	1	25
5006	Dhruv Shetty	M	2015-11-20	UI Developer	65000.00	8	2	28
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

● Find the employee whose designation is missing and fill it with "Data Scientist"

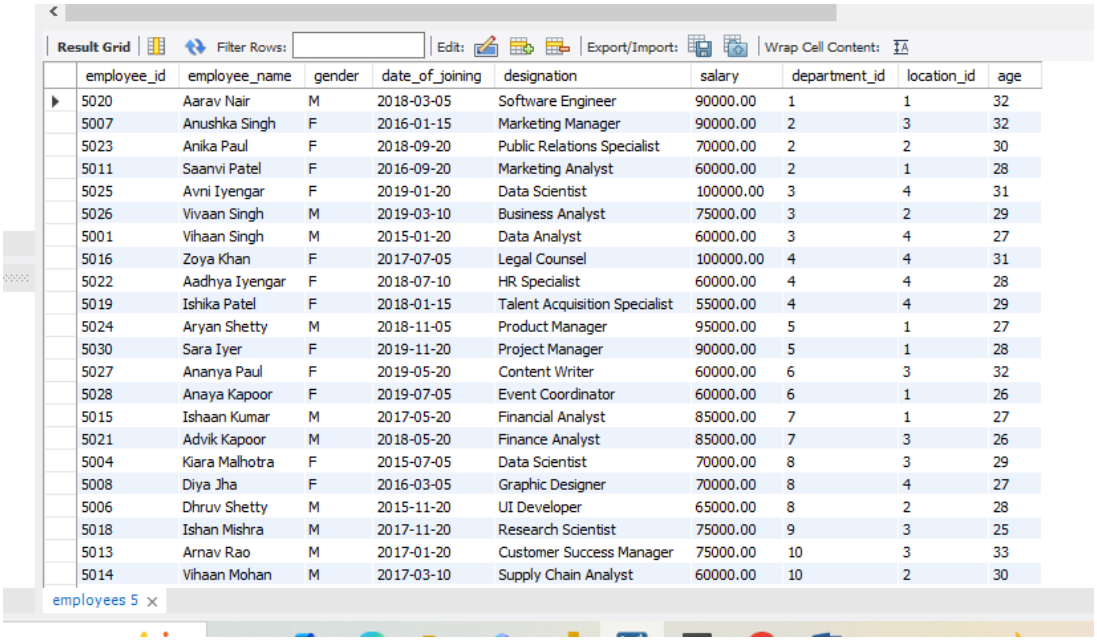


employee_id	employee_name	gender	date_of_joining	designation	salary	department_id	location_id	age
5001	Vihaan Singh	M	2015-01-20	Data Analyst	60000.00	3	4	27
5002	Reyansh Singh	M	2015-03-10	Network Engineer	80000.00	12	1	31
5003	Aaradhya Iyer	F	2015-05-20	Customer Support Executive	45000.00	10	2	26
5004	Kiara Malhotra	F	2015-07-05	Data Scientist	70000.00	8	3	29
5005	Anvi Chaudhary	F	2015-09-11	Business Development Executive	55000.00	11	1	25
5006	Dhruv Shetty	M	2015-11-20	UI Developer	65000.00	8	2	28
5007	Anushka Singh	F	2016-01-15	Marketing Manager	90000.00	2	3	32
5008	Diya Jha	F	2016-03-05	Graphic Designer	70000.00	8	4	27
5009	Kiaan Desai	M	2016-05-20	Sales Executive	55000.00	11	3	30
5010	Atharv Yadav	M	2016-07-10	Systems Administrator	80000.00	12	4	29
5011	Saanvi Patel	F	2016-09-20	Marketing Analyst	60000.00	2	1	28
5012	Myra Verma	F	2016-11-05	Operations Manager	95000.00	13	2	26
5013	Arnav Rao	M	2017-01-20	Customer Success Manager	75000.00	10	3	33
5014	Vihaan Mohan	M	2017-03-10	Supply Chain Analyst	60000.00	10	2	30
5015	Ishaan Kumar	M	2017-05-20	Financial Analyst	85000.00	7	1	27
5016	Zoya Khan	F	2017-07-05	Legal Counsel	100000.00	4	4	31
5017	Kabir Nair	M	2017-09-11	IT Support Specialist	80000.00	12	2	28
5018	Ishan Mishra	M	2017-11-20	Research Scientist	75000.00	9	3	25
5019	Ishika Patel	F	2018-01-15	Talent Acquisition Specialist	55000.00	4	4	29
5020	Aarav Nair	M	2018-03-05	Software Engineer	90000.00	1	1	32
5021	Advik Kapoor	M	2018-05-20	Finance Analyst	85000.00	7	3	26
5022	Aadhya Iyengar	F	2018-07-10	HR Specialist	60000.00	4	4	28

Sorting and Grouping Data:

1. ORDER BY:

● Find employees sorted by department ID in ascending order and salary in descending order



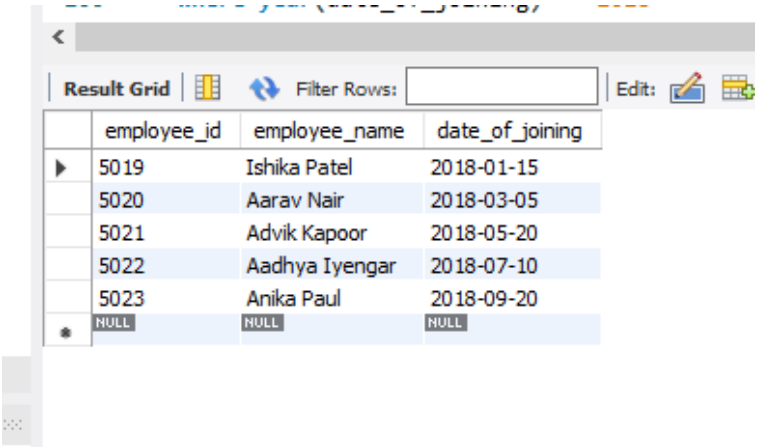
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	date_of_joining	designation	salary	department_id	location_id	age
5020	Aarav Nair	M	2018-03-05	Software Engineer	90000.00	1	1	32
5007	Anushka Singh	F	2016-01-15	Marketing Manager	90000.00	2	3	32
5023	Anika Paul	F	2018-09-20	Public Relations Specialist	70000.00	2	2	30
5011	Saanvi Patel	F	2016-09-20	Marketing Analyst	60000.00	2	1	28
5025	Avni Iyengar	F	2019-01-20	Data Scientist	100000.00	3	4	31
5026	Vivaan Singh	M	2019-03-10	Business Analyst	75000.00	3	2	29
5001	Vihaan Singh	M	2015-01-20	Data Analyst	60000.00	3	4	27
5016	Zoya Khan	F	2017-07-05	Legal Counsel	100000.00	4	4	31
5022	Aadhya Iyengar	F	2018-07-10	HR Specialist	60000.00	4	4	28
5019	Ishika Patel	F	2018-01-15	Talent Acquisition Specialist	55000.00	4	4	29
5024	Aryan Shetty	M	2018-11-05	Product Manager	95000.00	5	1	27
5030	Sara Iyer	F	2019-11-20	Project Manager	90000.00	5	1	28
5027	Ananya Paul	F	2019-05-20	Content Writer	60000.00	6	3	32
5028	Anaya Kapoor	F	2019-07-05	Event Coordinator	60000.00	6	1	26
5015	Ishaan Kumar	M	2017-05-20	Financial Analyst	85000.00	7	1	27
5021	Advik Kapoor	M	2018-05-20	Finance Analyst	85000.00	7	3	26
5004	Kiara Malhotra	F	2015-07-05	Data Scientist	70000.00	8	3	29
5008	Diya Jha	F	2016-03-05	Graphic Designer	70000.00	8	4	27
5006	Dhruv Shetty	M	2015-11-20	UI Developer	65000.00	8	2	28
5018	Ishan Mishra	M	2017-11-20	Research Scientist	75000.00	9	3	25
5013	Arnav Rao	M	2017-01-20	Customer Success Manager	75000.00	10	3	33
5014	Vihaan Mohan	M	2017-03-10	Supply Chain Analyst	60000.00	10	2	30

employees 5

2. LIMIT:

● Display the first 5 employees hired in the year 2018.



Result Grid | Filter Rows: | Edit:

employee_id	employee_name	date_of_joining
5019	Ishika Patel	2018-01-15
5020	Aarav Nair	2018-03-05
5021	Advik Kapoor	2018-05-20
5022	Aadhya Iyengar	2018-07-10
5023	Anika Paul	2018-09-20
NULL	NULL	NULL

3. AGGREGATE FUNCTIONS:

● Calculate the sum of all salaries in the Finance department.

```
107        join departments_info d on d.department_id = e.department_ic
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TotalFinance_Emp_Salary
170000.00

● Find the minimum age among all employees.

Result Grid | Filter Rows: | Edit:

employee_id	employee_name	age
5005	Anvi Chaudhary	25
5018	Ishan Mishra	25
NULL	NULL	NULL

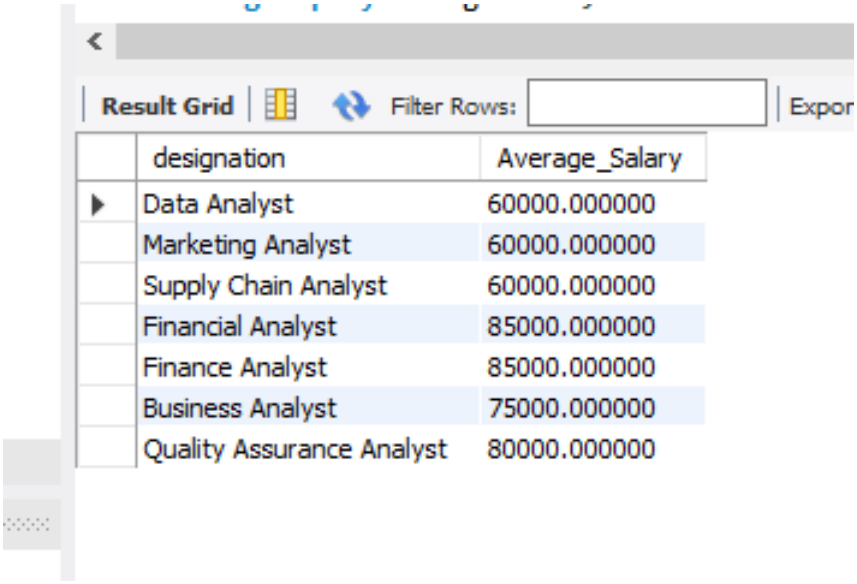
4. GROUP BY:

● List the maximum salary for each location

Result Grid | Filter Rows: | Export:

location_name	Maximum_Salary
Bangalore	95000.00
Chennai	95000.00
Hyderabad	90000.00
Pune	100000.00

● Calculate the average salary for each designation containing the word 'Analyst'

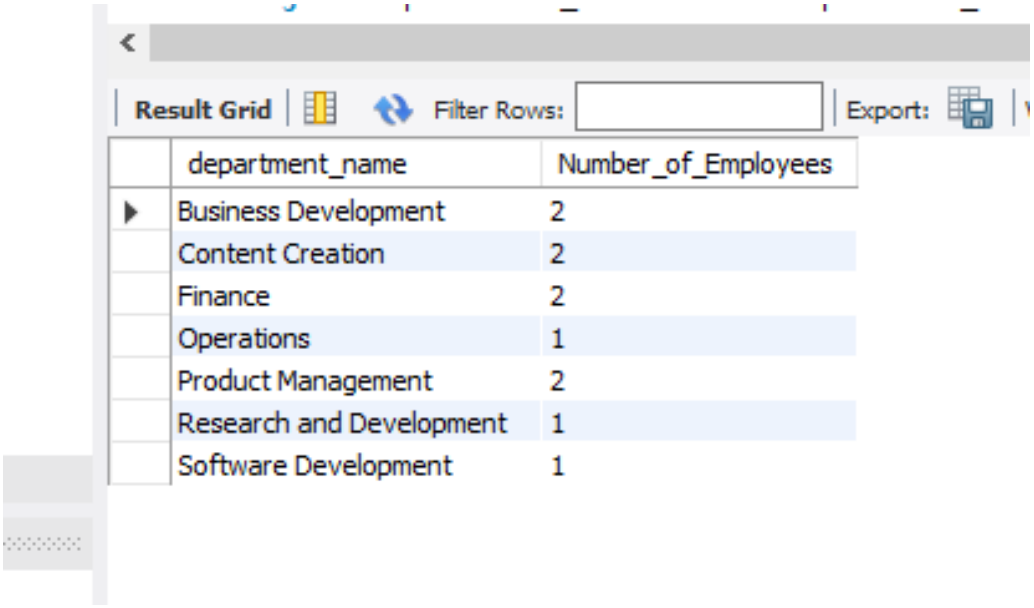


Result Grid | Filter Rows: | Expor

designation	Average_Salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING:

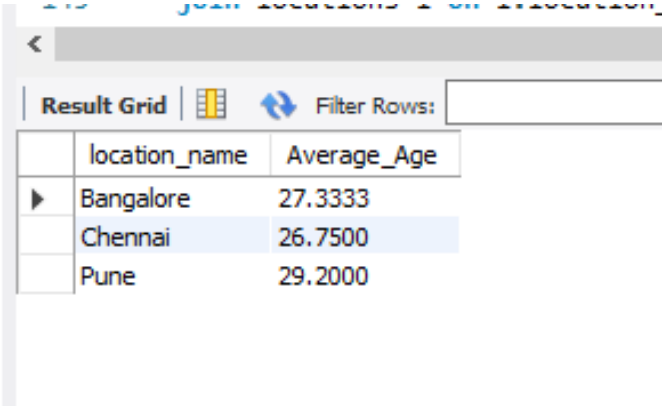
● Find departments with less than 3 employees.



Result Grid | Filter Rows: | Export:

department_name	Number_of_Employees
Business Development	2
Content Creation	2
Finance	2
Operations	1
Product Management	2
Research and Development	1
Software Development	1

● Find locations with female employees whose average age is below 30

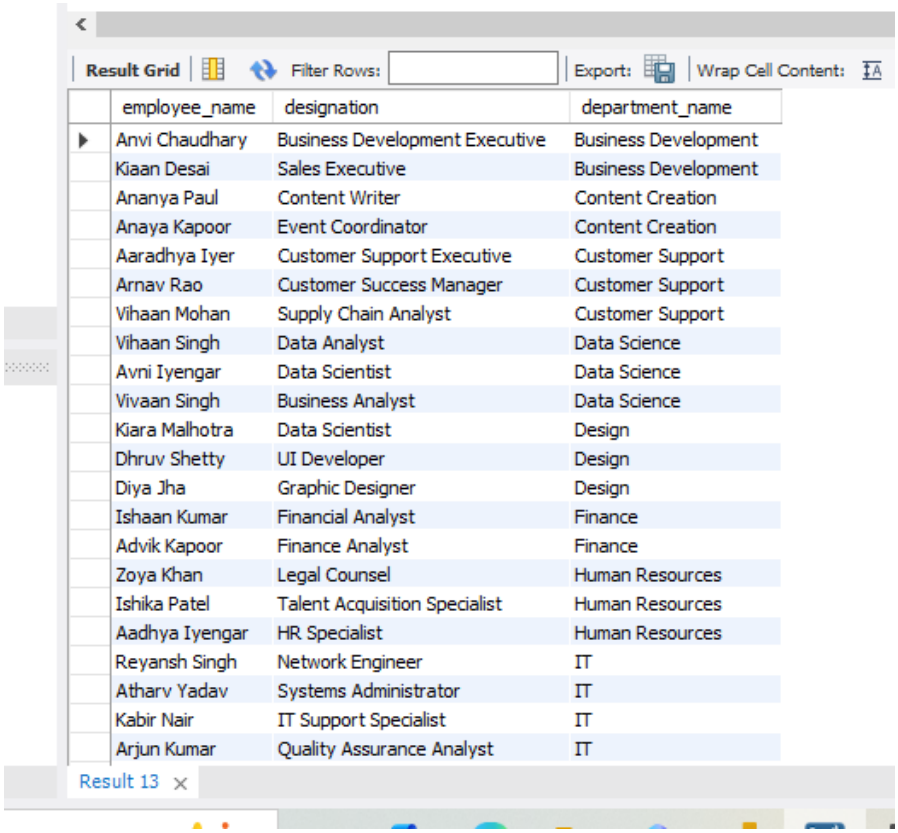


Result Grid | Filter Rows:

location_name	Average_Age
Bangalore	27.3333
Chennai	26.7500
Pune	29.2000

Joins:

1. **INNER JOIN:**
   ● List employee names, their designations, and department names where employees are assigned to a department.



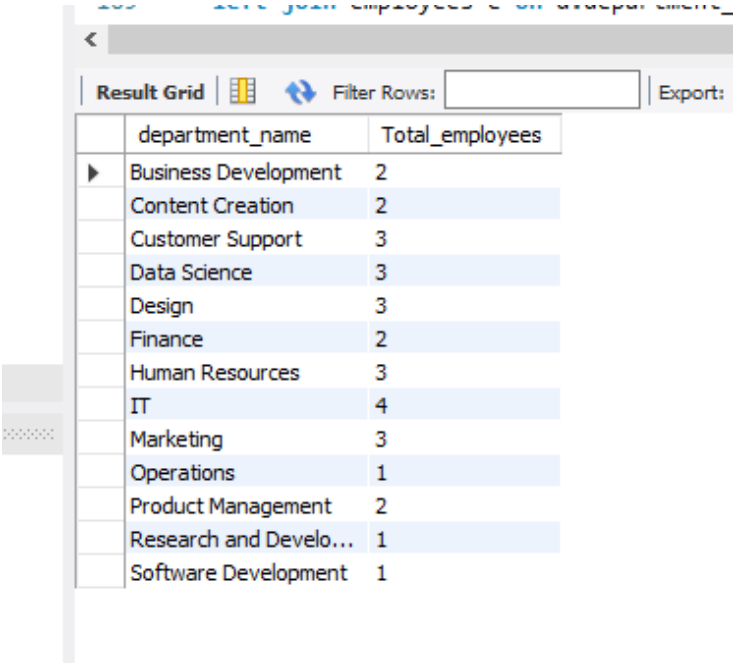
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science
Kiara Malhotra	Data Scientist	Design
Dhruv Shetty	UI Developer	Design
Diya Jha	Graphic Designer	Design
Ishaan Kumar	Financial Analyst	Finance
Advik Kapoor	Finance Analyst	Finance
Zoya Khan	Legal Counsel	Human Resources
Ishika Patel	Talent Acquisition Specialist	Human Resources
Aadhya Iyengar	HR Specialist	Human Resources
Reyansh Singh	Network Engineer	IT
Atharv Yadav	Systems Administrator	IT
Kabir Nair	IT Support Specialist	IT
Arjun Kumar	Quality Assurance Analyst	IT

Result 13

2. **LEFT JOIN:**

● List all departments along with the total number of employees in each department, including departments with no employees.

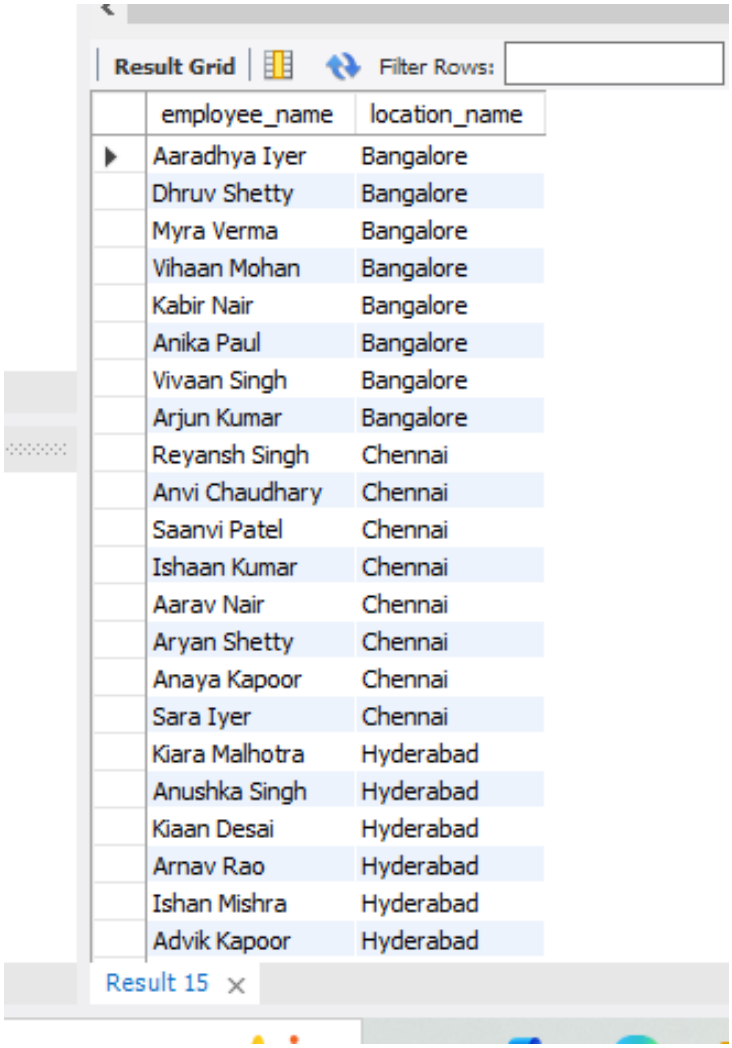


Result Grid | Filter Rows: | Export:

department_name	Total_employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Develo...	1
Software Development	1

3. **RIGHT JOIN:**

● Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name



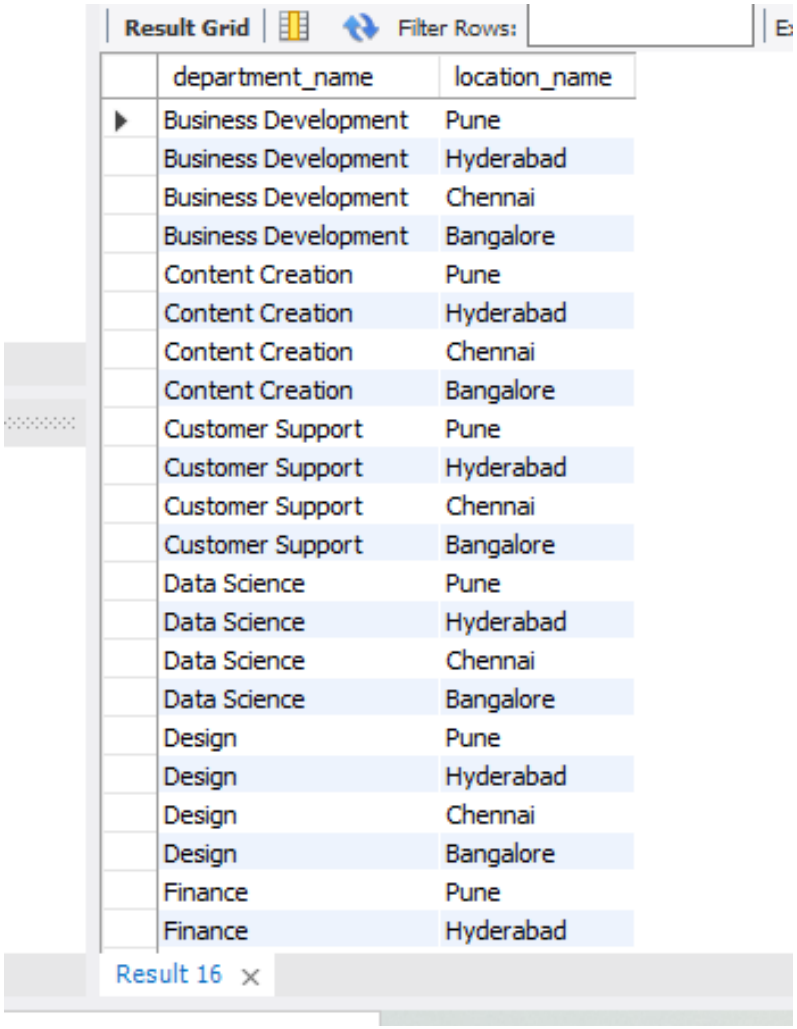
Result Grid | Filter Rows:

employee_name	location_name
Aaradhya Iyer	Bangalore
Dhruv Shetty	Bangalore
Myra Verma	Bangalore
Vihaan Mohan	Bangalore
Kabir Nair	Bangalore
Anika Paul	Bangalore
Vivaan Singh	Bangalore
Arjun Kumar	Bangalore
Reyansh Singh	Chennai
Anvi Chaudhary	Chennai
Saanvi Patel	Chennai
Ishaan Kumar	Chennai
Aarav Nair	Chennai
Aryan Shetty	Chennai
Anaya Kapoor	Chennai
Sara Iyer	Chennai
Kiara Malhotra	Hyderabad
Anushka Singh	Hyderabad
Kiaan Desai	Hyderabad
Arnav Rao	Hyderabad
Ishan Mishra	Hyderabad
Advik Kapoor	Hyderabad

Result 15

4. CROSS JOIN

- Show all possible combinations of departments and locations.



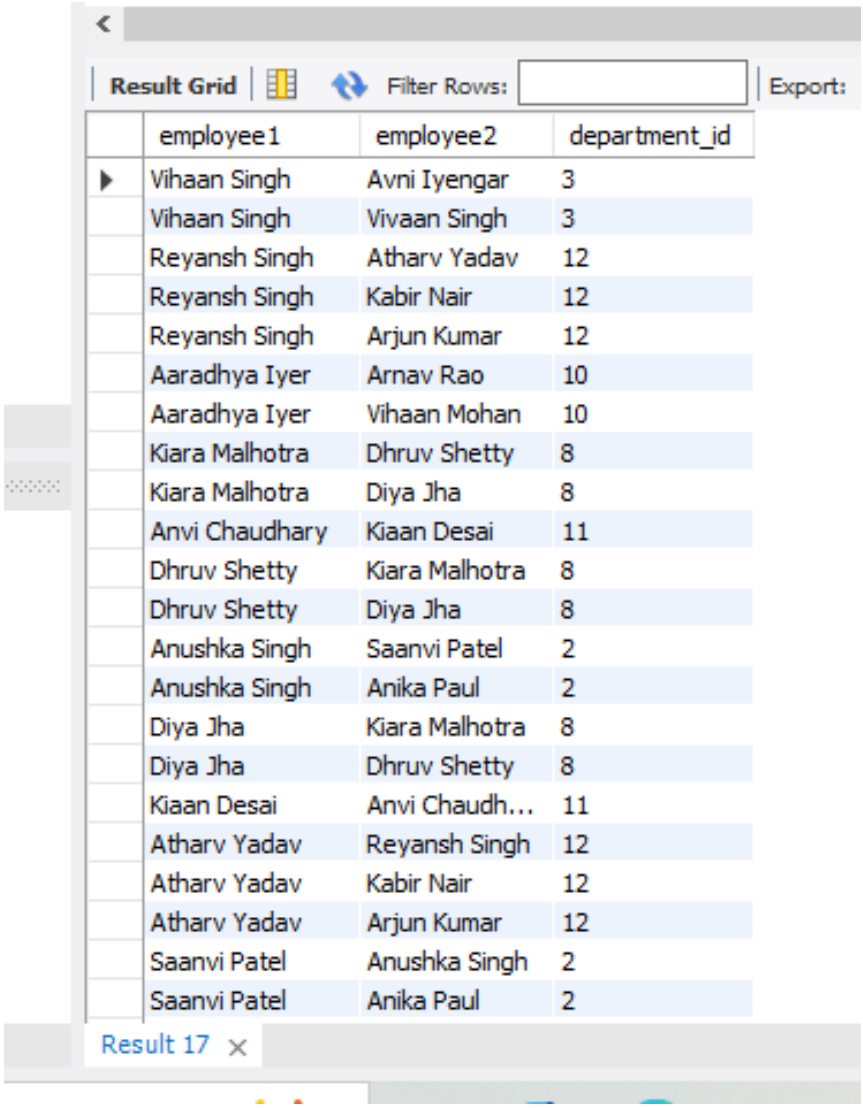
Result Grid | Filter Rows:

department_name	location_name
Business Development	Pune
Business Development	Hyderabad
Business Development	Chennai
Business Development	Bangalore
Content Creation	Pune
Content Creation	Hyderabad
Content Creation	Chennai
Content Creation	Bangalore
Customer Support	Pune
Customer Support	Hyderabad
Customer Support	Chennai
Customer Support	Bangalore
Data Science	Pune
Data Science	Hyderabad
Data Science	Chennai
Data Science	Bangalore
Design	Pune
Design	Hyderabad
Design	Chennai
Design	Bangalore
Finance	Pune
Finance	Hyderabad

Result 16

5. **SELF JOIN:**

● Show pairs of employees working in the same department, excluding self-pairs.



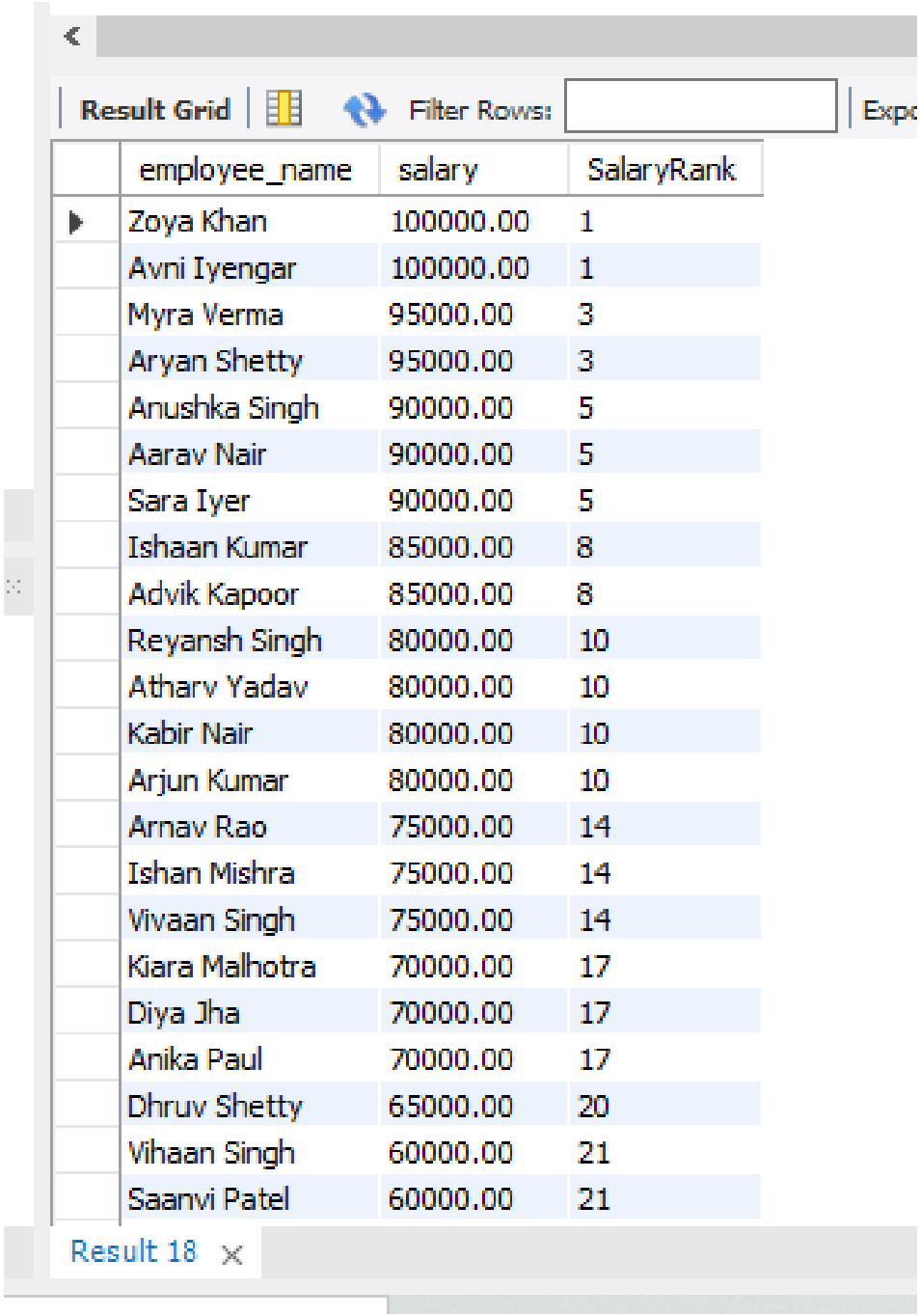
Result Grid | Filter Rows: | Export:

employee1	employee2	department_id
Vihaan Singh	Avni Iyengar	3
Vihaan Singh	Vivaan Singh	3
Reyansh Singh	Atharv Yadav	12
Reyansh Singh	Kabir Nair	12
Reyansh Singh	Arjun Kumar	12
Aaradhya Iyer	Arnav Rao	10
Aaradhya Iyer	Vihaan Mohan	10
Kiara Malhotra	Dhruv Shetty	8
Kiara Malhotra	Diya Jha	8
Anvi Chaudhary	Kiaan Desai	11
Dhruv Shetty	Kiara Malhotra	8
Dhruv Shetty	Diya Jha	8
Anushka Singh	Saanvi Patel	2
Anushka Singh	Anika Paul	2
Diya Jha	Kiara Malhotra	8
Diya Jha	Dhruv Shetty	8
Kiaan Desai	Anvi Chaudh...	11
Atharv Yadav	Reyansh Singh	12
Atharv Yadav	Kabir Nair	12
Atharv Yadav	Arjun Kumar	12
Saanvi Patel	Anushka Singh	2
Saanvi Patel	Anika Paul	2

Result 17

Windows function

- Write a window function query to rank employees by salary using rank()

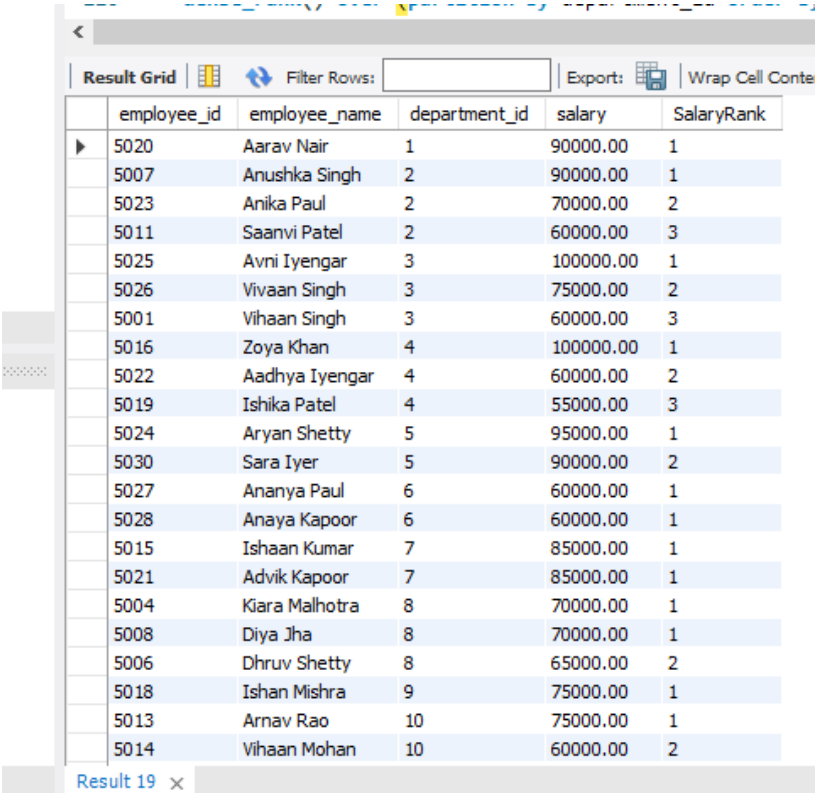


Result Grid | Filter Rows: | Expo

	employee_name	salary	SalaryRank
▶	Zoya Khan	100000.00	1
	Avni Iyengar	100000.00	1
	Myra Verma	95000.00	3
	Aryan Shetty	95000.00	3
	Anushka Singh	90000.00	5
	Aarav Nair	90000.00	5
	Sara Iyer	90000.00	5
	Ishaan Kumar	85000.00	8
	Advik Kapoor	85000.00	8
	Reyansh Singh	80000.00	10
	Atharv Yadav	80000.00	10
	Kabir Nair	80000.00	10
	Arjun Kumar	80000.00	10
	Arnav Rao	75000.00	14
	Ishan Mishra	75000.00	14
	Vivaan Singh	75000.00	14
	Kiara Malhotra	70000.00	17
	Diya Jha	70000.00	17
	Anika Paul	70000.00	17
	Dhruv Shetty	65000.00	20
	Vihaan Singh	60000.00	21
	Saanvi Patel	60000.00	21

Result 18

● Write a window function query to rank employees by salary within each department using DENSE_RANK()

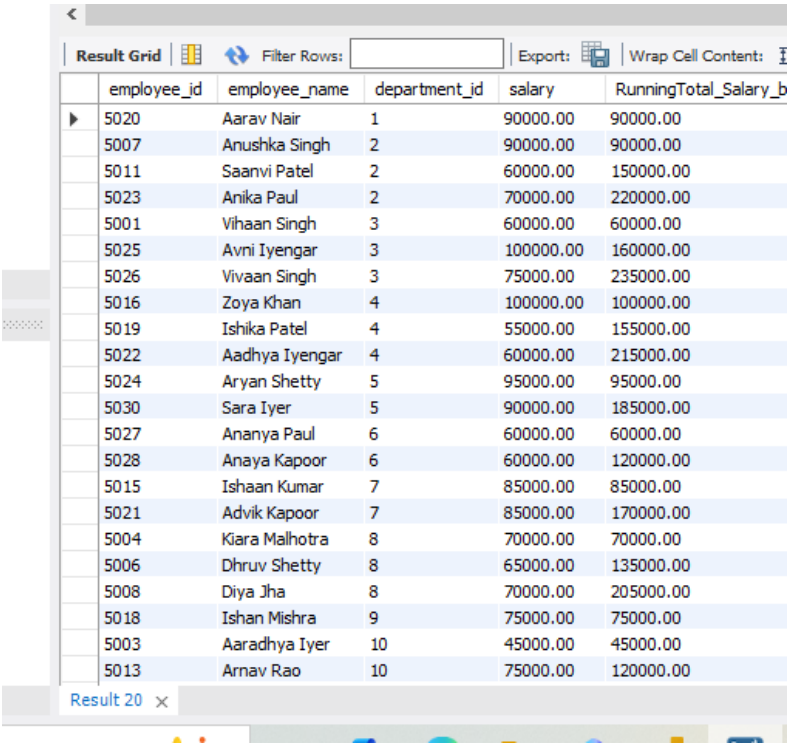


Result Grid | Filter Rows: | Export: | Wrap Cell Conte

employee_id	employee_name	department_id	salary	SalaryRank
5020	Aarav Nair	1	90000.00	1
5007	Anushka Singh	2	90000.00	1
5023	Anika Paul	2	70000.00	2
5011	Saanvi Patel	2	60000.00	3
5025	Avni Iyengar	3	100000.00	1
5026	Vivaan Singh	3	75000.00	2
5001	Vihaan Singh	3	60000.00	3
5016	Zoya Khan	4	100000.00	1
5022	Aadhya Iyengar	4	60000.00	2
5019	Ishika Patel	4	55000.00	3
5024	Aryan Shetty	5	95000.00	1
5030	Sara Iyer	5	90000.00	2
5027	Ananya Paul	6	60000.00	1
5028	Anaya Kapoor	6	60000.00	1
5015	Ishaan Kumar	7	85000.00	1
5021	Advik Kapoor	7	85000.00	1
5004	Kiara Malhotra	8	70000.00	1
5008	Diya Jha	8	70000.00	1
5006	Dhruv Shetty	8	65000.00	2
5018	Ishan Mishra	9	75000.00	1
5013	Arnav Rao	10	75000.00	1
5014	Vihaan Mohan	10	60000.00	2

Result 19

- Write a window function query, Running total salary by department



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_id	employee_name	department_id	salary	RunningTotal_Salary_b
5020	Aarav Nair	1	90000.00	90000.00
5007	Anushka Singh	2	90000.00	90000.00
5011	Saanvi Patel	2	60000.00	150000.00
5023	Anika Paul	2	70000.00	220000.00
5001	Vihaan Singh	3	60000.00	60000.00
5025	Avni Iyengar	3	100000.00	160000.00
5026	Vivaan Singh	3	75000.00	235000.00
5016	Zoya Khan	4	100000.00	100000.00
5019	Ishika Patel	4	55000.00	155000.00
5022	Aadhya Iyengar	4	60000.00	215000.00
5024	Aryan Shetty	5	95000.00	95000.00
5030	Sara Iyer	5	90000.00	185000.00
5027	Ananya Paul	6	60000.00	60000.00
5028	Anaya Kapoor	6	60000.00	120000.00
5015	Ishaan Kumar	7	85000.00	85000.00
5021	Advik Kapoor	7	85000.00	170000.00
5004	Kiara Malhotra	8	70000.00	70000.00
5006	Dhruv Shetty	8	65000.00	135000.00
5008	Diya Jha	8	70000.00	205000.00
5018	Ishan Mishra	9	75000.00	75000.00
5003	Aaradhya Iyer	10	45000.00	45000.00
5013	Arnav Rao	10	75000.00	120000.00

Result 20